VULLASA THRISHIKA

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence student at Amrita Vishwa Vidyapeetham, graduating in 2027, with research experience in biomedical signal processing at IIT (BHU). Hands-on experience developing AI/ML applications in Natural Language Processing, Generative AI, retrieval-augmented generation (RAG), and big data analytics. Skilled in Python and experienced with machine learning frameworks and data-processing technologies, with an emphasis on building practical AI solutions.

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore
B.Tech. in Artificial Intelligence | 2023–2027
CGPA: 6.5/10

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, MATLAB
AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision
Generative AI: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, FAISS, Hugging Face Transformers and Embeddings
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn
Big Data Technologies: Apache Spark, PySpark, Hadoop, Spark SQL, MapReduce
Tools & Platforms: Git, GitHub, Linux, Docker

RESEARCH EXPERIENCE

Research Intern — Biomedical Signal Processing May 2026 – June 2026
Indian Institute of Technology (BHU), Varanasi

- Analysed EEG-related data as part of biomedical signal processing research, including dataset analysis and computational pipeline development.
- Applied feature extraction and signal processing techniques, supported by experimental analysis.
- Documented methods and observations through literature review and research analysis.

PROJECTS

ColdChain Multi-AGV RL | *Python, PyTorch, DQN, Q-learning* | GitHub

- Developed a multi-agent warehouse simulation for cold-chain logistics, incorporating collision-aware navigation and thermal-risk modelling.
- Implemented reinforcement-learning and hybrid-control approaches to support warehouse task completion and risk-aware navigation.

MathForge AI | *Python, FastAPI, Rule-Based NLP* | GitHub

- Built an NLP pipeline that transforms natural-language mathematics problems into pseudocode, Python code, and computational complexity estimates.
- Developed algorithm classification and validation workflows to analyse algorithm patterns and time/space complexity.

Social-to-Lead Agentic AI | *Python, LangGraph, LangChain, FAISS, RAG* | GitHub

- Developed a conversational AI application for product assistance and lead qualification using retrieval-augmented generation (RAG).
- Integrated LangGraph workflows, FAISS vector search, and Hugging Face embeddings to support context-aware product information retrieval and multi-turn conversations.

Wikipedia Big Data Analytics | *Python, Hadoop, Spark, PySpark, Spark MLlib* | GitHub

- Developed a distributed data analytics pipeline using HDFS, Hadoop MapReduce, and Spark SQL to analyse Wikipedia article data.
- Applied Spark MLlib clustering and Python-based visualisation to explore article characteristics, categorisation patterns, and temporal trends.

HACKATHONS & COMPETITIONS

- KILN '26 — The Hybrid National Build Gauntlet | Cohere Live — Team thrishikav9
- Hack Devengers 2.0 | Devengers — Participant
- Quiz Competition | Devengers — Certificate of Achievement

CERTIFICATIONS

- GenAI Powered Data Analytics Job Simulation — Tata / Forage, 2026
- Introduction to Deep Learning — Infosys Springboard, 2026
- Introduction to Artificial Intelligence — Infosys Springboard, 2026
- Introduction to Data Science — Infosys Springboard, 2026